

Rotman

INTRO TO DATA VISUALIZATION

Part III Plotting Maps (with GeoPandas & Matplotlib) - Spatial data & GeoPandas basics

September 15, 2026 Prepared by Jay Cao / [MDAL](#)

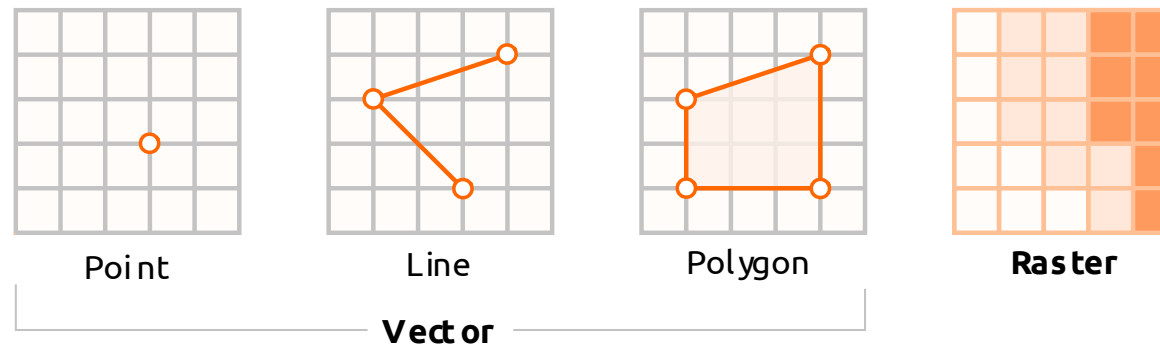
Website: <https://rmdal.github.io/mma-dv-2026/>



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Spatial Data

- A spatial dataset is a combination of...
 - location data and spatial dimensions (the where)
 - attribute data (the what)
- Two most common forms of spatial data
 - Vector
 - Raster



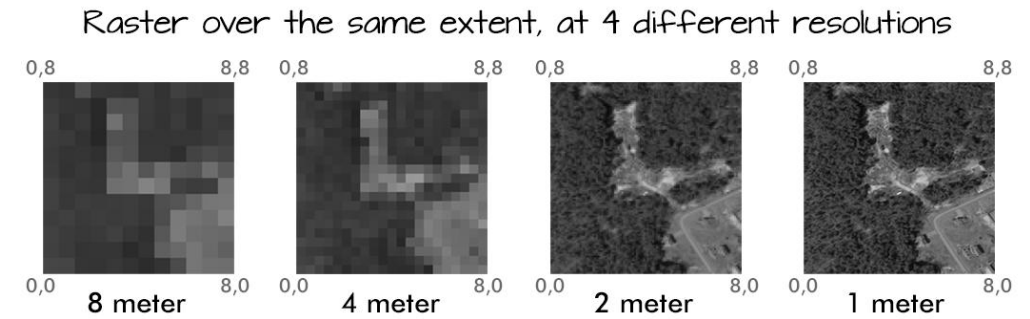
Vector Data

- Use geographic coordinates, or a series of coordinates, to create points, lines, and polygons representing real-world features
- Two most common vector data format
 - [GeoJSON](#) (a plain text file; .geojson, .json)
 - [Shapefile](#) (3 mandatory binary files + ...)
 - Shape (.shp) contains feature geometry
 - Shape index (.shx): facilitates fast search
 - Attribute (.dbf): contains attributes for each shape

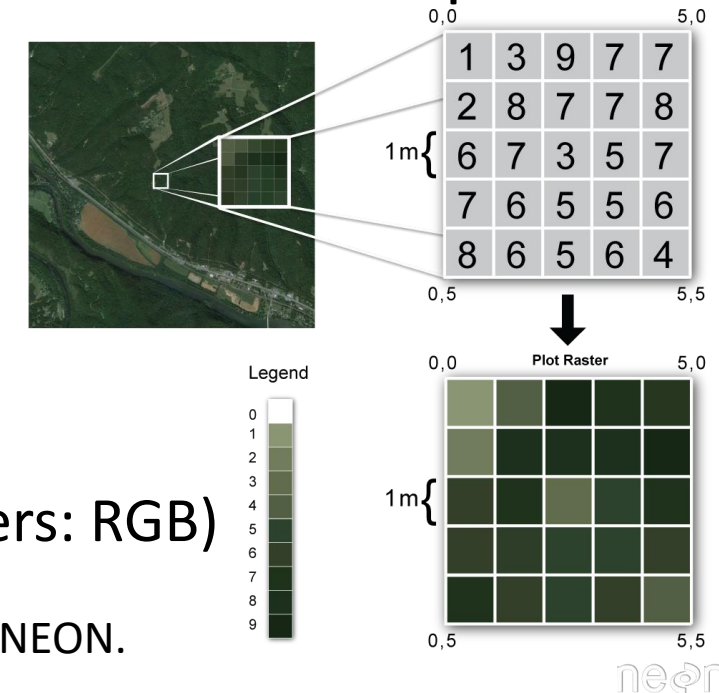
An example of GeoJSON file.

```
{
  "type": "Feature",
  "geometry": {
    "type": "Point",
    "coordinates": [125.6, 10.1]
  },
  "properties": {
    "name": "Dinagat Islands"
  }
}
```

Raster Data



- Represents space as a continuous grid with equal cell sizes
 - Each cell represents a square area (whose size depends on map resolution)
- Each cell contains a value pertaining to the type of feature it represents
 - Quantitative value (e.g., elevation, flood index)
 - Categorical value (e.g., type of land use)
- Examples of raster data (.tif, .jpeg, .png, etc.)
 - digital elevation models (DEMs)
 - Flood Susceptibility Index (FSI) map
 - satellite imagery (e.g., color image with 3 bands/layers: RGB)



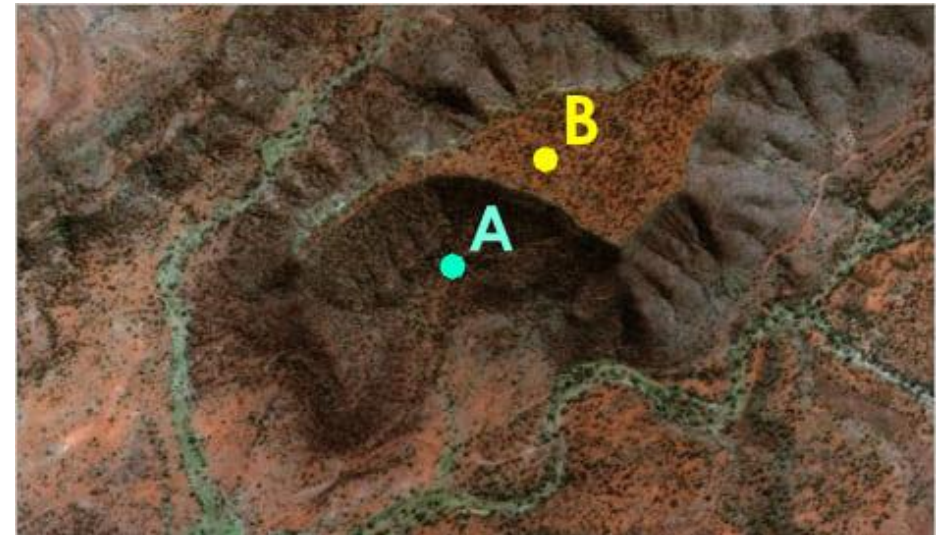
Source/Ref: 1) [Introduction to \(Q\)GIS](#) by [Jeff Allen](#); 2) [Raster 00 tutorial](#) from NEON.

Coordinate Reference System (CRS)

- Any geo-spatial dataset comes with a CRS
- Without CRS, one cannot plot and process geospatial data correctly
- CRS is “a framework used to precisely measure locations on the surface of Earth as coordinates.” ([wikipedia](#))
- CRS is a COMPLEX topic
 - Let’s attempt to understand some basics.

Motivation

- “URGENT: Rescue signal detected from satellite phone — coordinates 134.577° E, 24.006° S. Immediate response required.”
- Where is it on the Earth’s surface?
 - A or B?
 - GCS (Geographic Coordinate Systems; 3D)
- How to get there?
 - Need a map, which is 2D flat!
 - PCS (Projected Coordinate Systems; 3D- $\rightarrow$ 2D)

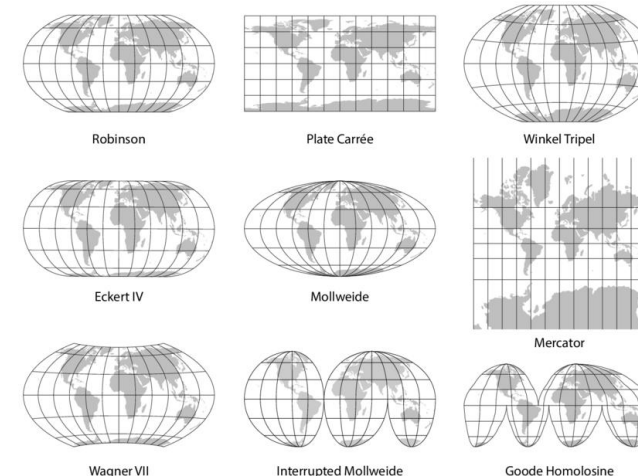
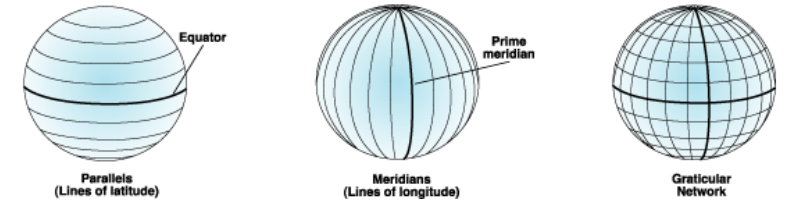
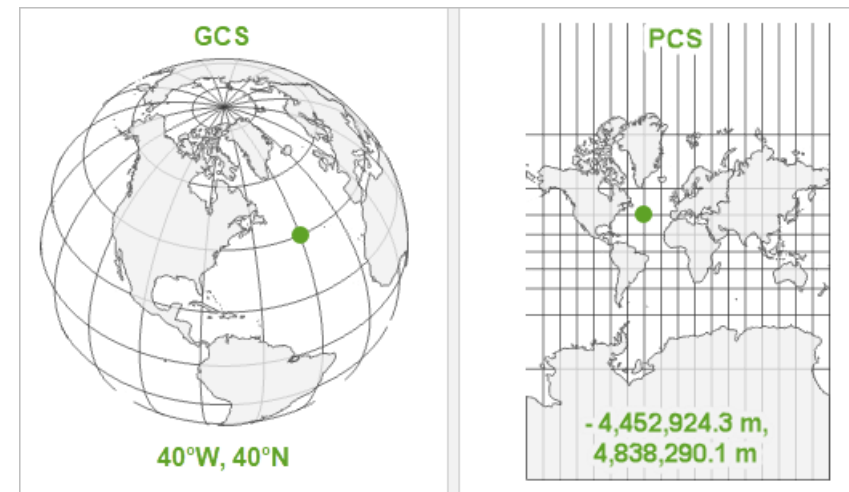


*A if GCS is Australian Geodetic Datum 1984
B if GCS is WGS (World Geodetic System) 1984*

Note: GCS is a specific type of CRS that model the Earth as a 3D sphere or ellipsoid, using angular units (like latitude and longitude)

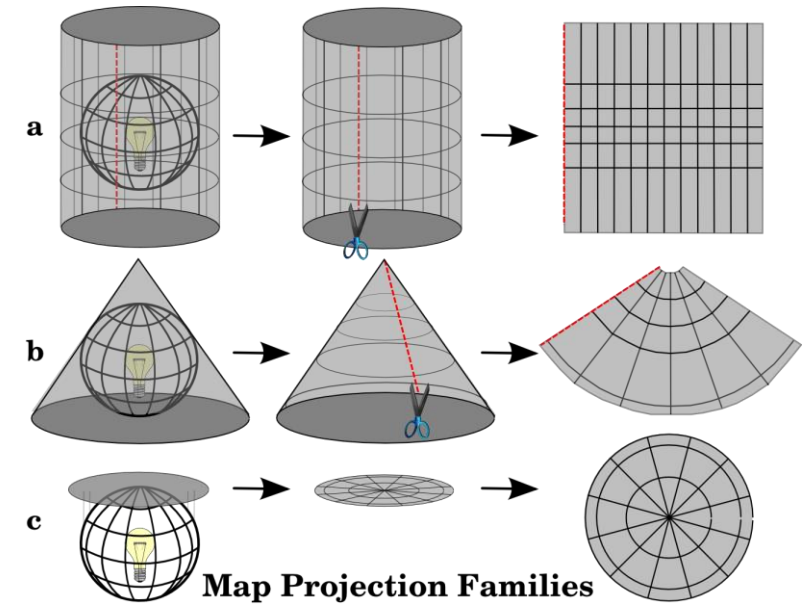
Two Main Types of CRS

- Geographic Coordinate Systems (GCS)
 - Round (where on the earth's surface)
 - Records locations in angular units (e.g., degrees)
 - Latitude: degrees north or south of equator
 - Longitude: degrees west or east of a prime meridian
 - Different models for the Earth surface -> *Many GCS*
 - e.g. World Geodetic System 1984 (WGS84)
- Projected Coordinate Systems (PCS)
 - Flat (how to draw on a 2d/flat paper)
 - Records locations in linear units (e.g., meters)
 - Many ways to project -> *Many PCS*
 - e.g., Universal Transverse Mercator (UTM)



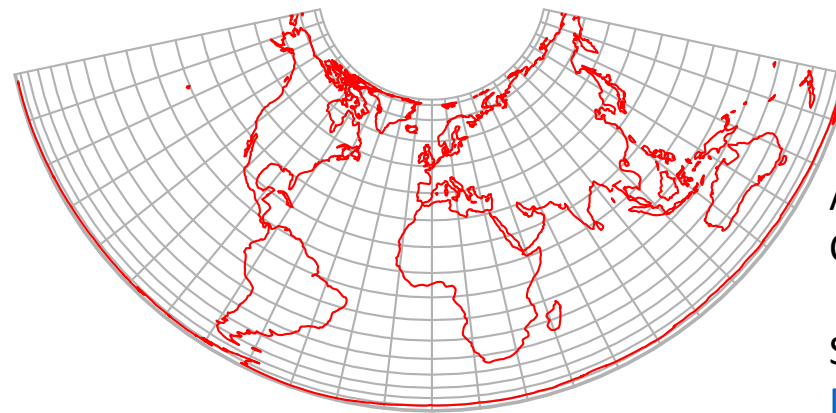
Projections

- MANY different projections
 - Three main families, but there are more
 - Within each family, there are many projections



Source: [QGIS Doc](#)

- A projection can preserve one or more properties below but never all
 - Direction
 - Distance
 - Area
 - Shape



Albers Equal-Area
Conic Projection

Source: [Wolfram MathWorld](#)

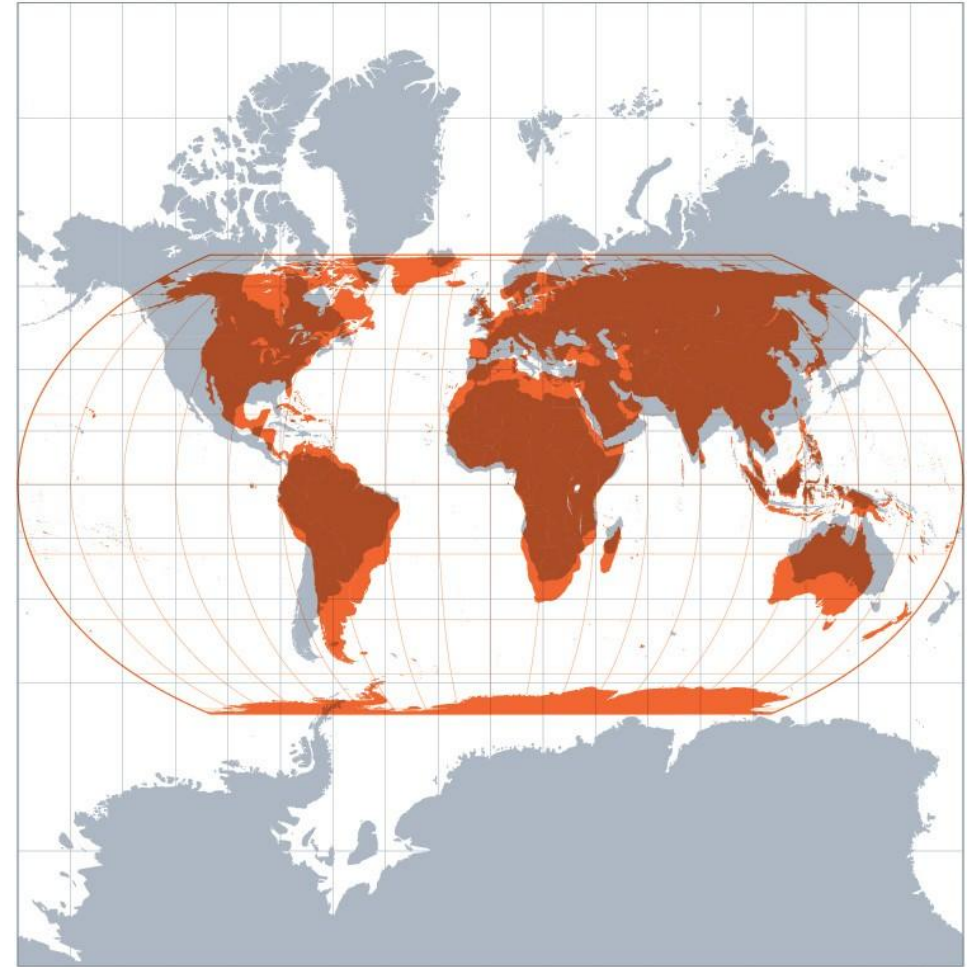
Projections: An Example

- “On 4 September [2026], the United Nations General Assembly voted to phase out use of the [Mercator](#) map, a sixteenth-century projection that is widely used but has been criticized for not representing countries’ sizes accurately.”
- “The UN resolution instead endorses use of the [Equal Earth](#) projection, which better shows the relative sizes of land masses.”

BENT INTO SHAPE

The Mercator projection keeps longitudinal lines straight — but the trade-off is that it gradually stretches land masses the closer they are to the poles. The curved meridians in the Equal Earth projection means that the relative size of land masses is depicted more accurately.

■ Mercator projection ■ Equal Earth projection



©nature

Source: [Hello Equal Earth: what the UN’s new world map will change](#) – Nature News

Good Read: [Explaining the true size of Africa, a lesson in map projections](#)

Components of CRS

- Coordinate system
 - E.g., longitude & latitude, x & y, etc.
- Datum: binds abstract coordinate system to real space on the Earth
 - Datum usually consists of
 - an estimate of the shape of the Earth (usually an ellipsoid)
 - one or more anchor points for which the measurement is documented
 - Examples of datum
 - Global datum: WGS84, North American Datum (NAD83)
 - Local datum: NAD27 (A local datum aligns its ellipsoid to closely fit the earth's surface in a particular area)
- Projection (if it is a PCS)

CRS is a “Stack” of Dependent Specifications

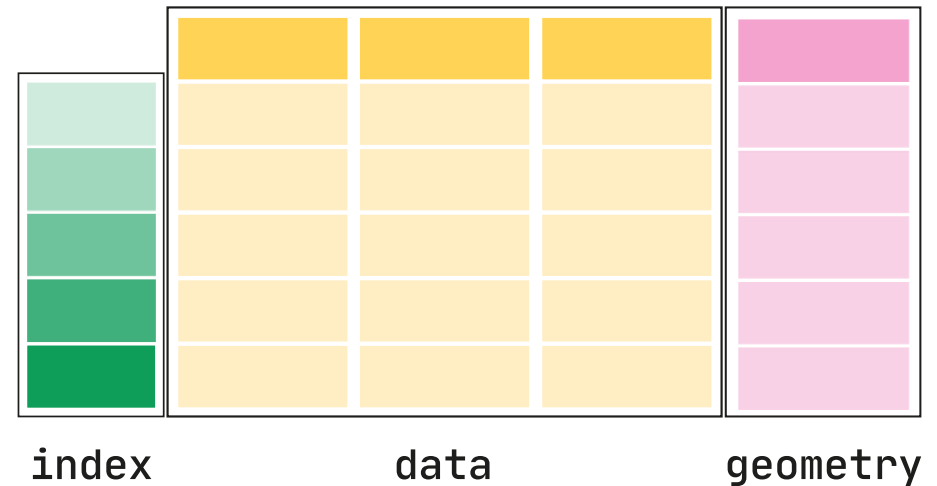
EPSG Code	Name	Ellipsoid	Horizontal Datum	CS Type	Projection	Origin	Axes	Unit of Measure
4326	GCS WGS 84	GRS 80	WGS 84	ellipsoidal (lat, lon)	N/A	equator/ prime meridian	equator, prime meridian	degree of arc
26717	UTM Zone 17N NAD 27	Clarke 1866	NAD 27	cartesian (x,y)	Transverse Mercator: central meridian 81°W, scaled 0.9996	500 km west of (81°W, 0°N)	equator, 81°W meridian	meter
6576	SPCS Tennessee Zone NAD 83 (2011) ftUS	GRS 80	NAD 83 (2011 epoch)	cartesian (x,y)	Lambert Conformal Conic: center 86°W, 34°20'N, standard parallels 35°15'N, 36°25'N	600 km grid west of center point	grid east at center point, 86°W meridian	US survey foot

GIS (Geographic Information Systems)

- GIS are tools to analyze, manipulate, and visualize spatial information on a computer
- Many GIS tools
 - [QGIS](#), [ArcGIS](#), [MapBox](#), etc.
- We will only focus on visualization with [GeoPandas](#)'s `plot()` function
 - GeoPandas's `plot()` is a method on `GeoSeries` or `GeoDataFrame`
 - GeoPandas's `plot()` builds on Matplotlib

Geopandas - 1

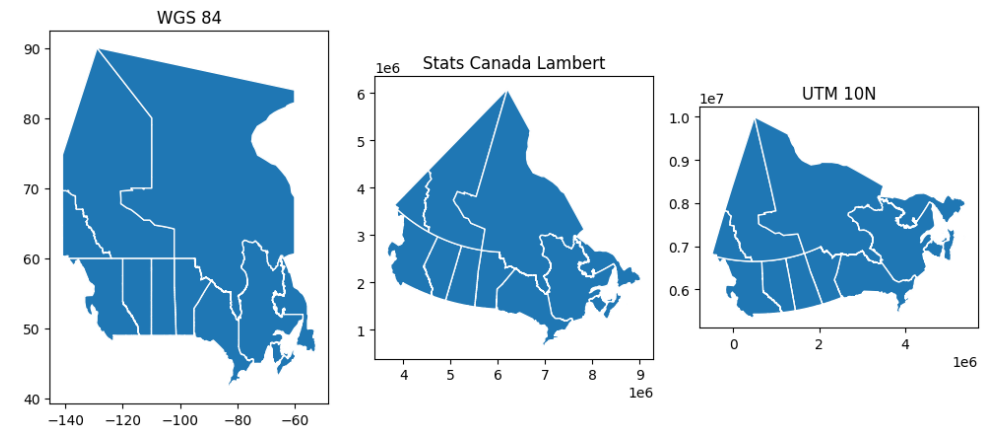
- Make working with geospatial data in Python easier
- Extends [pandas](#) to allow spatial operations on geometric types
 - Geometric operations built on [shapely](#)
 - File access built on [pyogrio](#)
 - Plotting built on [matplotlib](#)



Geopandas - 2

- Load dataset containing geometry info
- Manage Coordinate Reference System (CRS)
 - Store CRS info
 - translate between CRSs
- Geospatial operations
 - Calculate areas bounded by polygons
 - Spatial join
 - Spatial aggregation
 - Many more
- Plot

	PRENAME	PREABBR	LANDAREA	geometry
0	Newfoundland and Labrador	N.L.	3.581704e+05	POLYGON ((7644464.869 2980078.217, 7648864.423...
1	Prince Edward Island	P.E.I.	5.681179e+03	POLYGON ((8427184.671 1638777.314, 8427169.677...
2	Nova Scotia	N.S.	5.282471e+04	MULTIPOLYGON (((8693947.526 1494906.237, 86935...
3	New Brunswick	N.B.	7.124850e+04	POLYGON ((8188457.820 1707919.583, 8188444.194...
4	Quebec	Que.	1.298600e+06	MULTIPOLYGON (((8396476.571 1754341.151, 83956...
5	Ontario	Ont.	8.924118e+05	POLYGON ((6378273.940 2296884.400, 6378455.637...
6	Manitoba	Man.	5.403102e+05	POLYGON ((6039718.643 2636909.880, 6039717.720...
7	Saskatchewan	Sask.	5.770604e+05	POLYGON ((5248633.914 2767057.263, 5249285.640...
8	Alberta	Alta.	6.346583e+05	POLYGON ((5228304.177 2767597.891, 5228098.463...
9	British Columbia	B.C.	9.206866e+05	POLYGON ((4018904.414 3410247.271, 4019429.869...
10	Yukon	Y.T.	4.723454e+05	POLYGON ((4561932.471 4312865.174, 4564007.580...
11	Northwest Territories	N.W.T.	1.127712e+06	POLYGON ((5689672.257 4324508.314, 5685498.029...
12	Nunavut	Nvt.	1.836994e+06	POLYGON ((7297737.369 3983558.454, 7316653.440...



Hands-on: Warm-up – Geopandas Basics

- Load data
 - Canada province boundary shape file
- Determine CRS and convert between CRSs
- Perform geospatial operations
 - Example: calculating areas
- Plot maps